

Aim:

Write a Python program to calculate the sum of all natural numbers starting from 1 up to a given number n .
The program should:

- Accept a positive integer n from the user.
- If n is less than 1, display an appropriate message.
- Otherwise, compute the sum of numbers from 1 to n and display the result.

Input Format:

- The input should be a single positive integer entered by the user using the prompt:

Enter a positive integer:

This value defines the range (from 1 to n) for summation.

Output Format:

The output should display the sum of numbers from 1 to n .

- If $n \leq 0$, print:

Please enter a number greater than 0

- Otherwise, print the sum in the format:

The sum of numbers from 1 to $\langle n \rangle$ is: $\langle \text{result} \rangle$

Where $\langle \text{result} \rangle$ is the computed sum.

Source Code:

sum_upto_n.py

```
n=int(input("Enter a positive integer: "))
sum=0
i=0
if(n<=0):
    print("Please enter a number greater than 0")
else:
    while(i<=n):
        sum+=i
        i+=1
    print(f"The sum of numbers from 1 to {n} is: {sum}")
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter a positive integer:
1
The sum of numbers from 1 to 1 is: 1

Test Case - 2
User Output
Enter a positive integer:
100
The sum of numbers from 1 to 100 is: 5050

Test Case - 3
User Output
Enter a positive integer:
0
Please enter a number greater than 0

Test Case - 4
User Output
Enter a positive integer:
-5
Please enter a number greater than 0